Defect detection of automotive leather based on Nanodet-Plus

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***Abstract*—During the processing of automotive leather, various defects such as folding and oil stains can occur for several reasons. Therefore, it is essential to detect such defects before leather manufacturing to prevent them and enhance the quality of leather products. However, the current defect detection methods are not only expensive but also inefficient, indicating a need for a more efficient alternative. To address this problem, we constructed an Automotive-leather-defect dataset for vehicles based on collected leather defect samples. Subsequently, we proposed a defect detection method using the NanoDet-Plus neural network model, which can enhance the accuracy and speed of the detection process.**

Keywords—Defect detection, Automotive-leather-defect dataset, Convolutional Neural Network

I. INTRODUCTION

In the production of automotive leather, various factors can contribute to the occurrence of defects, such as folding and oil stains. Consequently, it is essential to detect such defects in the leather prior to manufacturing to prevent them and enhance the quality of the final leather products. The early detection of defects can also assist in reducing material and production costs. Therefore, the development of effective and efficient methods for defect detection is of significant interest to the automotive leather industry.

Currently, the detection of defects in leather is typically carried out by human inspectors, which has several limitations. This mode of operation is characterized by low detection efficiency and requires highly skilled testing personnel, resulting in the influence of human factors on the detection results, leading to potential inaccuracies in the detection results [1]. The advancement of computer vision detection technology and hardware conditions has led to increased interest in product defect detection using computer vision technology. Compared to traditional visual inspection methods, computer vision recognition can enhance the efficiency and accuracy of leather defect detection, reduce production costs, and improve the intelligence of leather processing [2]. Therefore, the application of computer vision technology in the leather industry has been widely explored by experts worldwide as a promising approach to enhance the efficiency and quality of the production process.

II. ANALYSIS OF OBJECT DETECTION METHODS BASED ON DEEP LEARNING

Visual inspection technology has witnessed significant development and implementation across various manufacturing industries. In the domain of leather defect detection based on visual inspection technology, experts at home and abroad have conducted research on leather defect visual detection algorithms and achieved promising results. For instance, Cui [3] proposed a leather defect detection method based on the Fuzzy C-means clustering algorithm (FCM), Yan [4] proposed a defect detection model based on Itti saliency calculation model, and Fan [5] presented a leather effective region extraction method based on saturation and global saliency. However, the above leather detection algorithms solely rely on traditional detection technology that is designed to target specific scenes or defect feature detection, thereby restricting their adaptability to multiple defect types and complex background textures of leather. As a result, their universality is limited. Hence, the development of more robust and versatile leather defect detection algorithms is a pressing research area.

Computer vision has emerged as a vital subfield of deep learning technology, and the development of convolutional neural network (CNN) models in 1998 has played a significant role in this area. In terms of defect detection algorithms, CNNs have demonstrated a high level of data expression ability. These models do not require the input of classification features and rely on the dataset to learn the relationship between features and their expression, enabling efficient and accurate detection of defects. Furthermore, the construction of CNN models is relatively simple, and they offer a high degree of accuracy in identifying defects. Hence, CNN-based defect detection algorithms are highly suitable for the efficient and accurate detection of leather defects.

Deep learning-based object detection algorithm has two technical routes: Anchor based(one-stage, two-stage) and Anchor free. Dense anchor boxes are generated by the anchor mechanism, so that the network can directly carry out target classification and bounding box regression on this base. two-stage algorithm firstly selects the target to be detected and then classifies and regression. Representative algorithms include RCNN, Fast RCNN, Faster RCNN, etc. The cloth surface defect detection method based on Cascade RCNN proposed by Xiong [6] and the leather surface defect detection based on Faster RCNN proposed by Fang [7] have achieved great results. Later, in order to solve the problem of slow detection speed of two-stage algorithm, Redmon et al. [8] proposed a one-stage algorithm(yolo), which integrated the functions of target detection, positioning and classification into one network. The leather defect detection method based on improved yolov3 proposed by Tang [9] and the fabric defect small target detection method based on improved SSD algorithm proposed by Zhao [10] have greatly improved the leather defect detection effect.

In recent times, there has been a growing interest in anchor-free technology, which eliminates the need for anchor boxes and performs detection through keypoint estimation, reducing the number of network hyperparameters. In 2019, Tian Z et al. [11] introduced the Fully Convolutional One-Stage Object Detection algorithm as a novel approach to

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object detection in computer vision. The algorithm employs a fully convolutional network, and keypoint estimation is used to detect and locate objects. This technique has shown great potential and is a promising alternative to anchor-based methods in object detection.

NanoDet-Plus [12] is a one-stage Anchor-free object detection model based on convolutional neural network, which follows the FCOS architecture. The model is highly efficient in terms of performance and has lightweight characteristics, making it easy to train and deploy on mobile devices. As a result, NanoDet-Plus is an optimal solution for detecting leather defects with low costs and high detection requirements.

III. Algorithm Implementation

To enhance the accuracy and efficiency of the leather defect detection, the NanoDet-Plus network model is chosen and utilized for the deep training of the leather image dataset. The network model is fine-tuned by adjusting the structure and parameters of the detection network, and the performance of the model is evaluated through several rounds of iteration. Through these iterations, an optimized detection model with improved accuracy and faster detection speed is obtained.

A. Introduction of NanoDet-plus

The structure diagram of NanoDet-Plus leather defect detection model is shown in Fig 1.

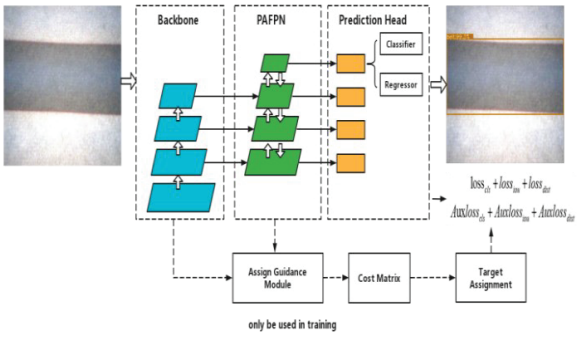


Fig. 1. Structure diagram of NanoDet-plus

$loss_{cls}$ is Quality Focal Loss(QFL) in Generalized Focal Loss [13]. $loss_{dist}$ refers to Distribution Focal Loss(DFL) of Generalized Focal Loss. $loss_{iou}$ is the Generalized IoU Loss [14]. $Auxloss_{cls}$, $Auxloss_{dist}$ and $Auxloss_{iou}$ are the losses corresponding to the Assign Guidance Module (AGM).

B. Label Matching Policy

Object detection model training requires assigning labels to the objects in an image, and the label assignment process is a critical step in achieving good performance. Initially, object detection models used direct location-based matching to assign labels, but this method has limitations. Later, the anchor-iou matching method was developed, which is widely used in current object detection models. In recent years, dynamic matching strategies based on Matching Cost have been proposed, such as the Hungarian matching algorithm used in the DETR model [15], which have further improved the performance of object detection models. The evolution of label assignment strategies has contributed significantly to the progress of object detection technology.

The NanoDet-plus model employs only two Depthwise separable convolution modules for simultaneous classification and regression prediction. Due to its random initialization state, it is not suitable for the computation of Match Cost for matching. To overcome this limitation, the model utilizes Label Assignment Distillation (LAD) and knowledge distilling [16], relying on the label predictions of the teacher's network to guide student network training. To address the challenge of optimal label matching in a lightweight model with a dynamic soft label assignment strategy, the authors introduce an auxiliary module called Assign Guidance Module (AGM). The AGM module features four 3×3 normal convolutions and utilizes Group Normalization as the Normalize layer. Furthermore, it shares parameters in Feature maps at different scales, and is only used during training, which does not impact the reasoning speed in any way.

C. Backbone

The ShuffleNetV2 model [17] is employed as the Backbone Network to extract features that have been downsampled by factors of 8, 16, and 32, which are subsequently fed into the Path Aggregation Network (PAN) for multi-scale feature fusion.

D. PAFPN

Adapting the cross-stage partial (CSP) path aggregation network (PAN) feature fusion module from YOLOv5 [18], the NanoDet-Plus model incorporates a lightweight GhostPAN module.

The GhostBlock module from GhostNet [19] is utilized in the GhostPAN module to handle feature fusion across multiple layers. GhostPAN consists of a basic building block that includes a collection of 1×1 convolutions and 3×3 Depthwise separable convolutions, resulting in a very small number of parameters and computations required.

E. Prediction Head

Due to the fact that Batch Normalization can merge its normalized parameters directly into the convolution during inference, the normalization method used in the detection head has been changed from Group Normalization to Batch Normalization.

NanoDet-Plus adopts the FCOS Head consisting of four branches, each with 256 channels. To reduce computational burden and enhance efficiency, depthwise separable convolution is used instead of ordinary convolution. The kernel size of the depthwise separable convolution is changed to 5×5, and the number of convolution stacks is reduced from four to two groups, with the number of channels decreased from 256 dimensions to 96 dimensions. As a result, border regression and classification can be carried out by the same set of convolution, streamlining the computation.

F. Loss Function

Given the complexity of the background of leather defects and the category imbalance, the classification and bounding box regression are conducted using Generalized Focal Loss, which comprises of Quality Focal Loss (QFL) and Distribution Focal Loss (DFL). This loss function aims to address the problem of class imbalance and emphasize the importance of the difficult samples during training.

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Specifically, QFL reduces the weight of well-classified samples and emphasizes the importance of misclassified samples by introducing a quality factor, whereas DFL reduces the weight of easy samples and assigns more weight to the difficult ones by introducing a distribution factor. By using this loss function, NanoDet-Plus can achieve better performance in the detection of leather defects.

Quality Focal Loss(QFL): Classification-IoU joint representation is proposed based on Focal Loss [20]. The Classification score is no longer the classification score, but the IoU score, the label $y \in \{0,1\}$ discrete value becomes continuous value between $y \in [0,1]$. Therefore, as in (1), Quality Focal Loss is the expansion form of Focal Loss on continuous labels:

$$QFL(\sigma) = -\left|y-\sigma\right|^{\beta}((1-y)\log(1-\sigma)+y\log(\sigma)) \quad (1)$$

Where the quality label $y \in [0,1]$, σ is the prediction, the global minimum solution of QFL is then $y=\sigma$, β is the regulator.

Distribution Focal Loss(DFL): Generalized Focal Loss uses an arbitrary probability distribution for bounding frame location and reduces the required complete probability distribution to the probabilities of two distance points relative to the true coordinates. As in (2), the corresponding loss function is:

$$DFL(S_i, S_{i+1}) = -((y_{i+1}-y)\log(S_i)+(y-y_i)\log(S_{i+1})) \quad (2)$$

Where S_i represents the probability that the regression distance is y_i . The meaning is to optimize the probability of the left and right positions closest to the label in a form similar to cross entropy, so that the network can quickly focus on the distribution of the adjacent regions of the target location.

G. Training Strategy

The selected optimizer is AdamW [21], which is less sensitive to hyperparameters and has faster convergence compared to other optimizers. The learning rate reduction strategy employed is CosineAnnealingLR [22], and gradient clipping is added to the back-propagation calculation of the gradient. Additionally, Exponential Moving Average (EMA) [23] is used as a model smoothing strategy, which is a popular technique in deep learning.

IV. Make the Defect Dataset

Deep learning algorithms often require abundant data support in the training process, and a larger dataset can improve the performance and generalization ability of the trained network. The quality and complexity of the dataset can have a significant impact on the effectiveness of the deep learning network. In the case of detecting leather defects, the complexity of the images, as well as the variety of defect categories and shapes, can pose challenges to the model. Thus, it is crucial to construct a defect dataset for vehicle composite leather that is reflective of the actual production process in order to train the model effectively.

A. Image Acquisition

The experimental setup for image acquisition is illustrated in Fig 2. The sample is placed on an acrylic plate which acts as a support platform. The height of the platform can be adjusted using four column feet. The light source is provided by three horizontally placed 7.2W adjustable light tubes positioned at 5-7cm intervals beneath the platform. The images of the defective area are captured using a 1080p camera as the image acquisition unit.



Fig. 2. Image acquisition system

Some sample images of leather defects collected by the image acquisition system are shown in the Fig 3:

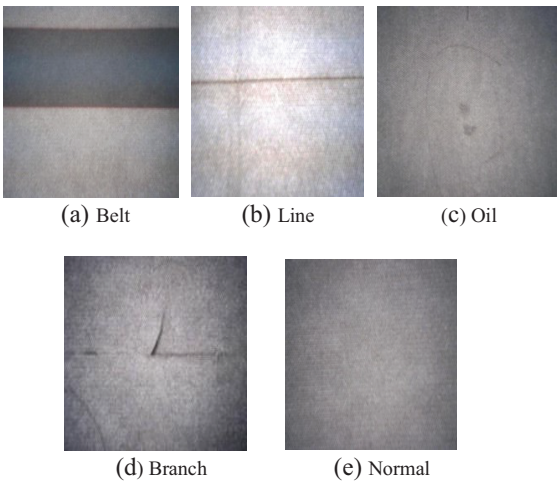


Fig. 3. Samples of defects in each category

As depicted in Fig 3, panel (a) displays a defect resembling a band, which is annotated as Belt. Panel (b) illustrates a defect that appears linear, labeled as Line. Panel (c) shows an oil-based defect, marked as Oil. Panel (d) features a defect with a tree-like appearance, annotated as Branch. Finally, panel (e) represents a regular sample with no defects, labeled as Normal.

B. Data Preprocessing

During the image acquisition process, various factors such as variations in lighting, inaccuracies in the image acquisition equipment, and noise during image transmission can result in suboptimal image quality in datasets. Since the quality of the image can significantly impact the effectiveness of detection algorithms, it is necessary to perform preprocessing on the images before feature extraction to mitigate such issues.

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To standardize the input data for neural network training, two preprocessing steps were taken on the leather defect dataset. Firstly, images were cropped to a size of 1500×1080 in order to minimize the impact of surrounding shadows. Secondly, the images in the dataset had different sizes, which is not suitable for training a neural network model. Therefore, to address this issue, the size of the images was standardized to 416×416.

C. Data Augmentation

In deep learning, having a large amount of data is crucial for a model to achieve high performance and generalization ability. However, in some cases, the available data may be insufficient. In order to address this issue, data augmentation can be used to increase the amount of data for training. Data augmentation typically involves applying image transformations such as resizing, flipping, rotating, and adjusting color values such as hue, saturation, and brightness to generate new training samples. In the context of constructing the leather defect dataset, data augmentation techniques such as flipping, random rotation, and adjustments to hue, saturation, and brightness were used to expand the dataset and increase the variety of samples available for training.

After annotating the images with LabelImg tool, a total of 894 defect samples were obtained, comprising 114 belt defect samples, 192 line defect samples, 132 mixed strip and linear defect samples, 180 oil stain samples, 132 tree defect samples, and 144 normal samples. The labeled images were then divided into three sets, namely training set, validation set, and test set, in an 8:1:1 ratio.

V. Experiment

This study employs the NanoDet-Plus neural network model, utilizing the Pytorch deep learning framework for model training. The training is conducted on a single-card RTX3080, with a batch-size of 32 and 300 iterations. To minimize hardware expenses in the future, tensorRT is implemented to enhance inference on the GTX1660ti device, and the trtexec tool is utilized to assess the network's performance.

A. Performance Test

Various lightweight backbone networks, including ShuffleNetV2, MobileNetV2 [24], EfficientNet-lite [25], and RepVGG [26], were chosen for multiple rounds of training. The models that best met the task requirements were selected through experimental evaluations.

Finally, the performance of different Backbone models on the test set is shown in TABLE 1:

TABLE I. Performance Comparison Of Different Backbones

Backbone	Metrics			
	mAP (0.5:0.95)	*Latency*	*Params*	*Size*
ShffleNetV2	0.75	3.33ms	4.17M	16.2M
MobileNetV2	0.75	4.37ms	5.73M	22.3M
EfficientNet-lite	0.76	4.57ms	9.52M	36.8M
RepVGG	0.75	4.12ms	12.2M	47.1M

Compared to other lightweight networks, such as MobileNetV2, the ShuffleNetV2 backbone network model has fewer parameters, lower computational complexity, and smaller weight size while maintaining similar accuracy. Thus, in the following experiments, ShuffleNetV2 is chosen as the backbone network.

The loss curve and mAP (0.5:0.95) curve when ShuffleNetV2 is used as the backbone network model for training are shown in Fig 4 and Fig 5:

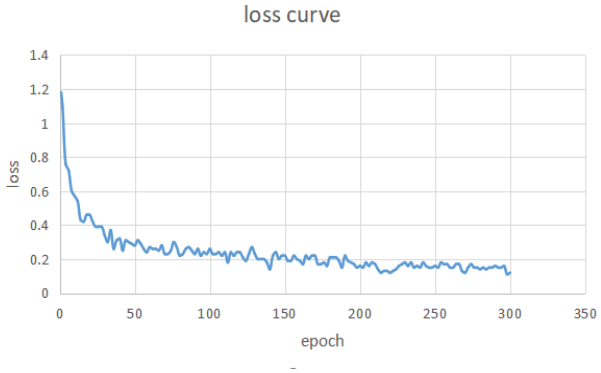


Fig. 4. Loss curve of during model training

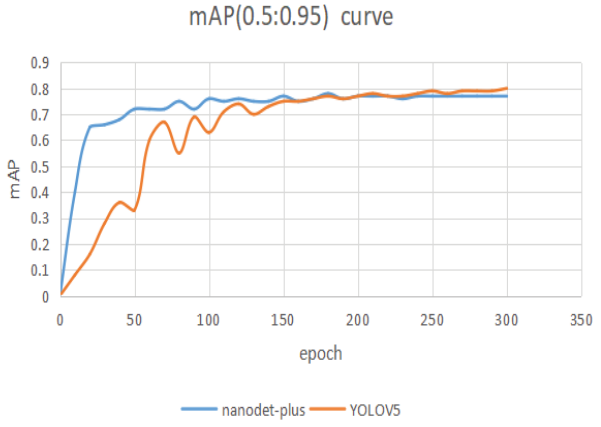


Fig. 5. mAP curve during model training

When the IoU threshold is set to 0.45 and the confidence threshold is set to 0.25, the confusion matrix of each category is presented in Fig 6:

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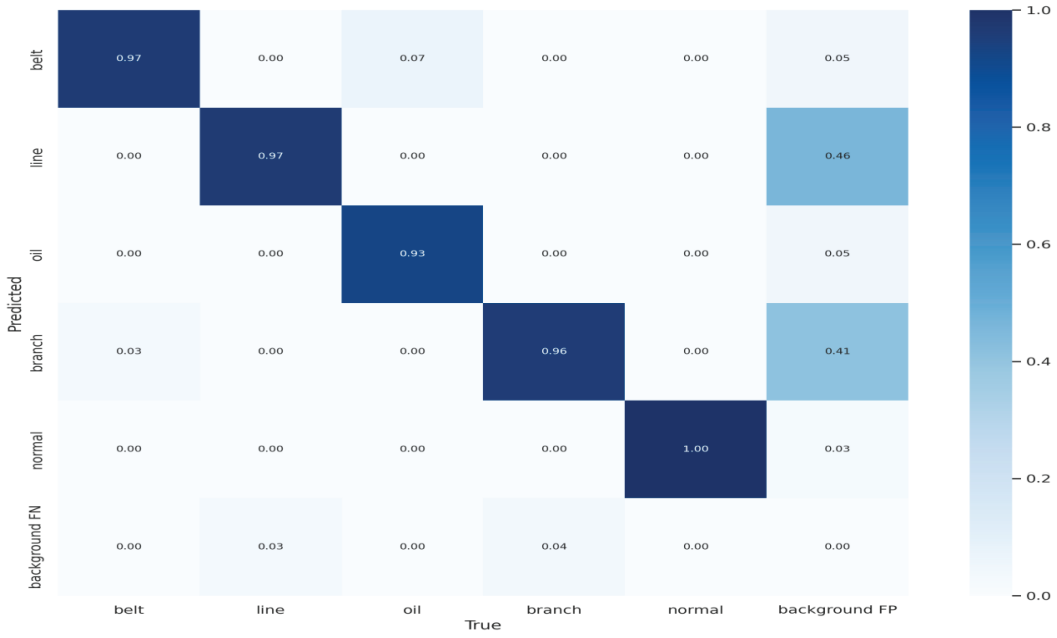


Fig. 6. Confusion matrix of each category

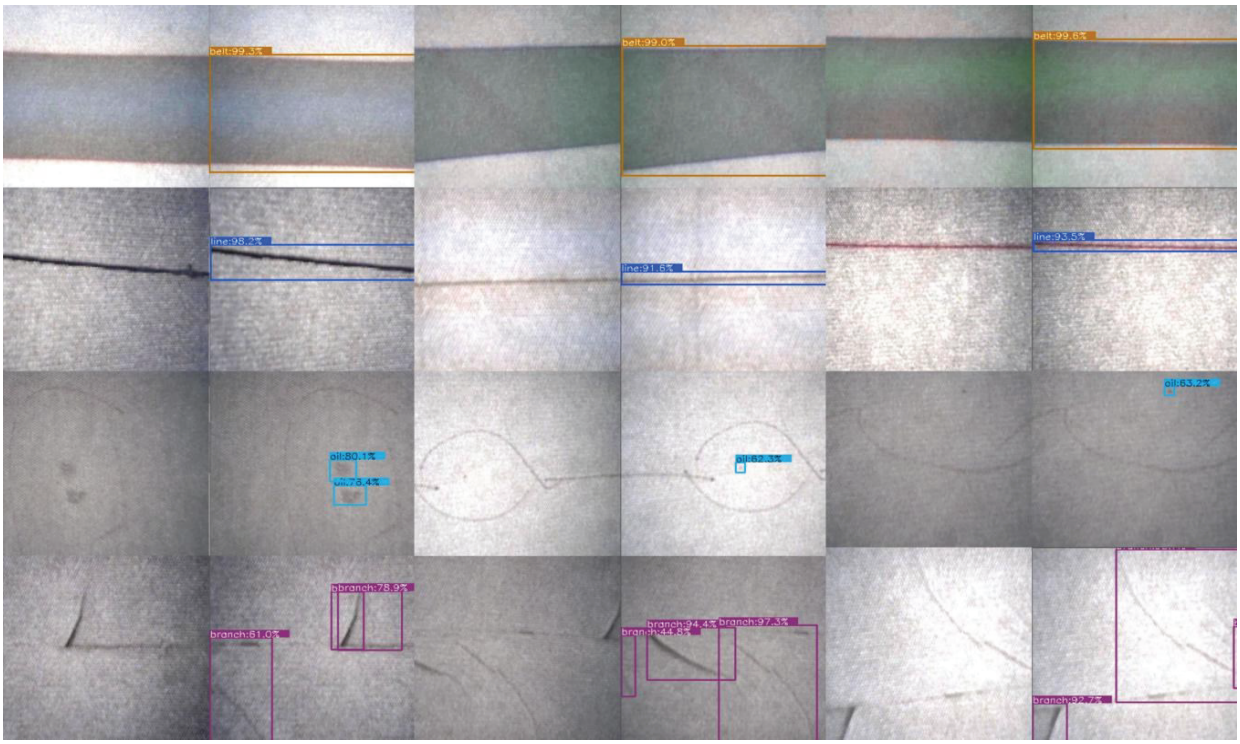


Fig. 7. Defect detection results

As presented in Fig 6, the NanoDet-Plus model designed for leather defect detection exhibits high accuracy in identifying different defect types. Nonetheless, the model shows some limitations in differentiating between background and line/branch defects. Further analysis confirmed that line and branch defects share similarities with the background in terms of brightness and contrast within a certain range, which may account for these misidentifications.

Fig 7 presents the defect detection results of each model for each defect type. The first defect type is belt defect, the second is line defect, the third is oil stain defect, and the fourth is branch defect. The model exhibits a high level of confidence in its predictions for the belt and line classes, with a confidence score of over 90%. However, its predictions for the oil and branch classes are slightly less confident, with a confidence score of over 60%.

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The results show that NanoDet-Plus successfully detects all defects in the test samples with high confidence, demonstrating its efficacy in meeting the requirements of leather defect detection in industrial production.

B. Performance Comparison

TABLE II. PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	Metrics			
	mAP(0.5:0.95)	***Latency***	***Params***	***Size***
Nanodet [12]	0.68	3.53ms	0.95M	7.73M
Nanodet-plus	0.75	3.33ms	4.17M	16.2M
Yolov5 [18]	0.80	8.0ms	7.2M	13.8M
Yolov7 [27]	0.81	29.6ms	36.9M	71.3M

As shown in Tabel II, compared to the NanoDet detection model, the NanoDet-Plus model achieves a 7% improvement in mean average precision (mAP) while maintaining similar inference delay. However, the YOLOv5s detection model exhibits a 5% higher mAP than NanoDet-Plus, but at the cost of a 140% increase in inference delay. Similarly, the YOLOv7 model achieves only a 5% improvement in mAP over NanoDet-Plus, but with a significant 788% increase in inference delay.

In consideration of the requirement to reduce costs in practical applications, it may be necessary to deploy the leather defect detection model to mobile devices with limited computing power. Based on our findings, NanoDet-Plus is better suited for this task in industrial production due to its superior performance in terms of both accuracy and efficiency.

VI. CONCLUSION

In this paper, we employ the NanoDet-Plus object detection model to detect defects on automotive leather. Through experimental comparisons, it is found that the NanoDet-Plus model achieves a balance between detection speed and accuracy in the field of automotive leather defect detection, making it suitable for industrial production. The algorithm has low hardware requirements, making it cost-effective for mass production. However, some problems still need to be addressed, such as the impact of data set processing on detection results, the limited number of data samples, and difficulties distinguishing between similar ribbon and linear defects, which can affect detection performance. These areas require further investigation and improvement.

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